



HNDIT4242-Computer Services Management



Lecture 08 Availability Management



Lesson Objectives

- Define **availability** in IT service management.
- Explain key **availability metrics** (Uptime, Downtime, MTBF, MTTR).
- Understand **redundancy, failover, and disaster recovery** strategies.
- Learn how **proactive maintenance** improves availability.
- Explore **real-world examples** of high-availability systems.



1. Understanding Availability in IT Services

1.1 What is IT Service Availability?

- Availability refers to the ability of a system, service, or infrastructure to remain operational and accessible to users.
- It is a key factor in **Service Level Agreements (SLAs)**.
- The goal is to **minimize downtime** and maximize uptime.



1.2 Importance of High Availability in IT Services

- High availability is critical for:
 - ✓ Online businesses (e-commerce, banking, streaming services).
 - ✓ Cloud computing and SaaS applications.
 - ✓ Enterprise IT systems (ERP, CRM, internal networks).



1.3 How Downtime Impacts Businesses

- **Lost revenue:** An e-commerce website going offline results in direct sales loss.
- **Customer dissatisfaction:** Frequent downtime leads to user frustration.
- **Security risks:** System failures can expose vulnerabilities to cyberattacks.



1.4 Key Availability Metrics

These metrics help IT teams **measure and improve** availability.

Metric	Definition
Uptime (%)	The percentage of time a system is operational.
Downtime	The amount of time a system is unavailable.
Mean Time Between Failures (MTBF)	The average time between system failures.
Mean Time to Repair (MTTR)	The average time required to restore a system after failure.



2. Techniques to Ensure High Availability

- ***2.1 Redundancy and Failover Systems***

- **Redundancy:** Having backup components (servers, storage, power supply) ready to take over in case of failure.
- **Failover Systems:** Automatic switching to a backup system when a failure occurs.
- **Load Balancing:** Distributing network or application traffic across multiple servers to avoid overload.

Examples:

- ✓ **Google Cloud, AWS, and Microsoft Azure** use multiple data centers for redundancy.
- ✓ **Banking systems** ensure 24/7 availability by implementing failover mechanisms.



2.2 Disaster Recovery (DR) and Backup Strategies

- **Disaster Recovery Plan (DRP):** A structured approach to restoring services after a major failure (e.g., cyberattack, hardware failure, natural disaster).
- **Backup Strategies:**
 - **Full backup:** Copies all data and files.
 - **Incremental backup:** Saves only changes since the last backup.
 - **Cloud-based backup:** Uses online storage solutions for offsite data protection.
- **Example:** Banks and healthcare systems use **geographically dispersed data centers** to ensure service continuity.



2.3 Proactive Maintenance for Availability

- **Regular system updates and patches** prevent vulnerabilities.
- **Monitoring tools** (e.g., Nagios, Zabbix) detect potential failures before they happen.
- **Predictive analytics** uses AI to anticipate hardware failures.



Summary

- **Availability management** ensures IT services remain operational with minimal downtime.
- **High availability** is critical for businesses, financial institutions, and cloud services.
- **Metrics like MTBF and MTTR** help measure and improve system reliability.
- **Redundancy, failover systems, and backups** prevent service disruptions.
- **Proactive maintenance** helps detect and fix issues before they cause downtime.



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